Module – 4:
Raspberry Pi Programming

Introduction

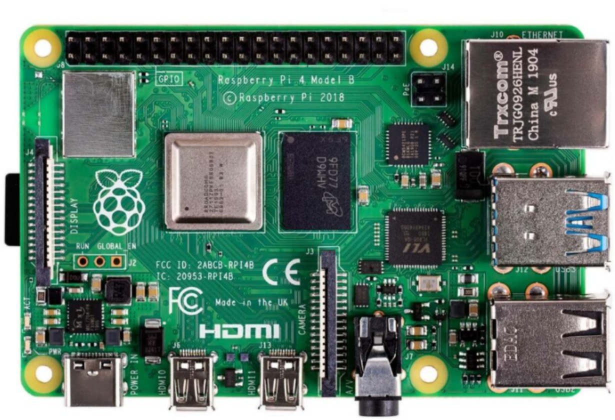
Simple Definition

A **Raspberry Pi** is a **credit-card-sized computer** that can:

- Run an operating system (like Linux)
- Connect to keyboard, mouse, monitor
- Perform tasks like a normal PC

What can you do with it?

- Mini desktop computer
- CCTV / NVR system
- Web server
- Robotics controller
- IoT gateway
- Retro gaming console



What makes it special?

1. Complete Computer on a Board

It includes:

- Processor (CPU)
- RAM
- USB ports
- HDMI output
- GPIO pins (for hardware interfacing)

No bulky CPU cabinet needed

2. GPIO Pins (Key Feature)

- Used to connect sensors, motors, LEDs
- Makes it ideal for:
  - IoT projects
  - Robotics
  - Automation

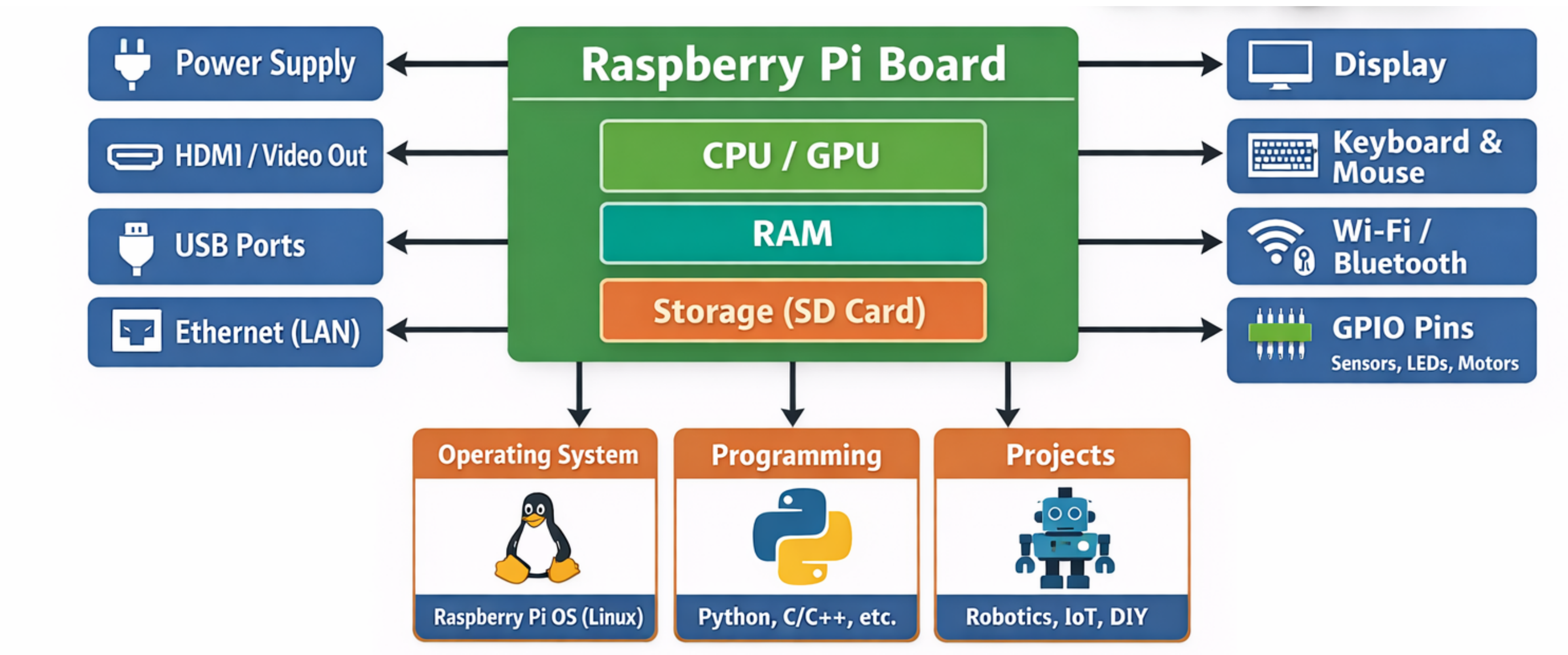
3. Runs Real Operating System

- Typically uses **Raspberry Pi OS (Linux)**
- Supports:
  - Python
  - C/C++
  - Java

4. Low Cost

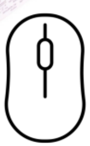
- Much cheaper than a traditional computer
- Designed for education and hobby projects

Raspberry Pi – General Overview

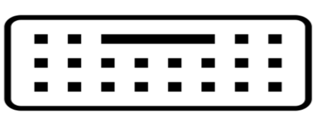


USB C **Power**
15W (5V, 3A)

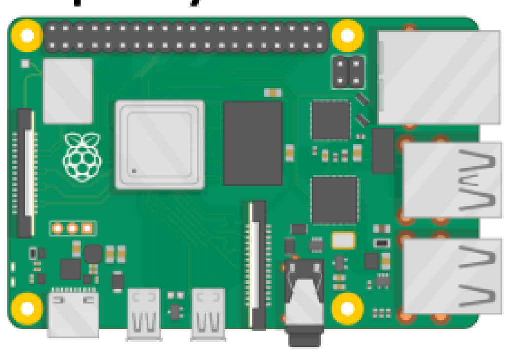
Mouse



Keyboard



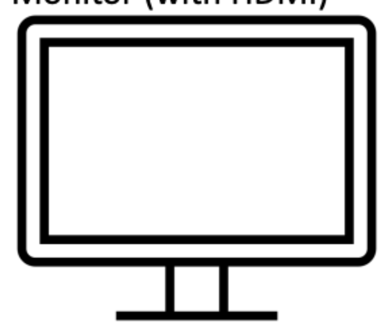
Raspberry Pi



Micro SD Card (+ Adapter)

(8GB or more)

Monitor (with HDMI)

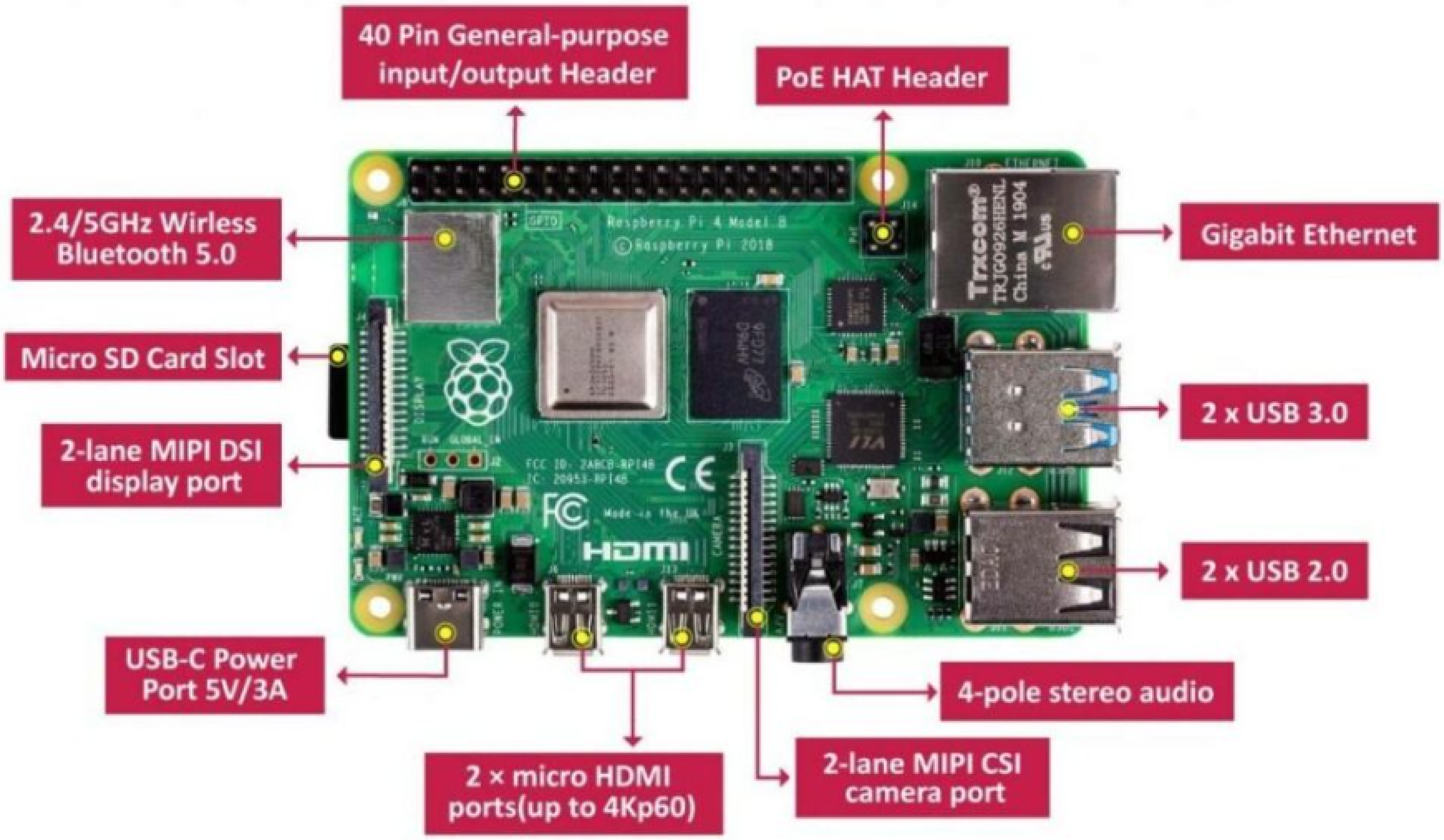


Raspberry Pi 4 - Features

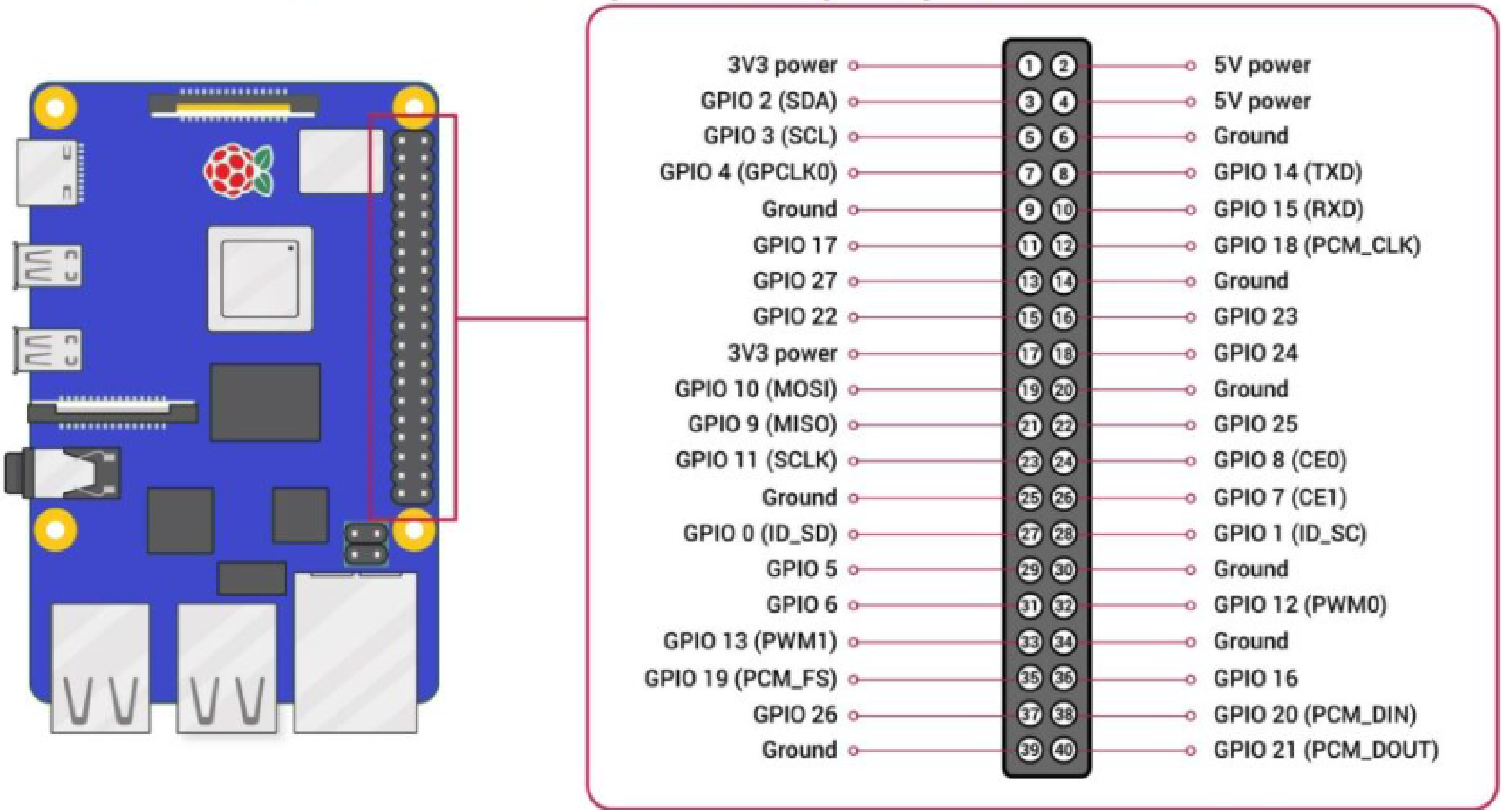
Quad-core 1.5 GHz ARM Cortex-A72 CPU
→ Provides faster processing for multitasking and applications
2GB / 4GB / 8GB RAM options
→ Supports better performance for different workload requirements
Dual 4K display support
→ Can drive two monitors for desktop and multimedia use
Video Core VI GPU
→ Enables smooth graphics and video playback
2× USB 3.0 + 2× USB 2.0 ports
→ High-speed connectivity for peripherals
Gigabit Ethernet → Ensures fast and reliable wired networking
Dual-band Wi-Fi & Bluetooth 5.0
→ Provides wireless connectivity for IoT and communication
USB Type-C power (5V, 3A)
→ Stable and efficient power supply
40-pin GPIO header
→ Allows interfacing with sensors, motors, and external devices
MicroSD card support
→ Used for OS installation and data storage
CSI camera & DSI display ports
→ Supports camera modules and touchscreen displays
Improved architecture over previous models
→ Offers better speed, connectivity, and overall performance

Raspberry Pi 4 Model

Raspberry Pi 4 – Hardware View



Raspberry Pi 4 – Pin Details

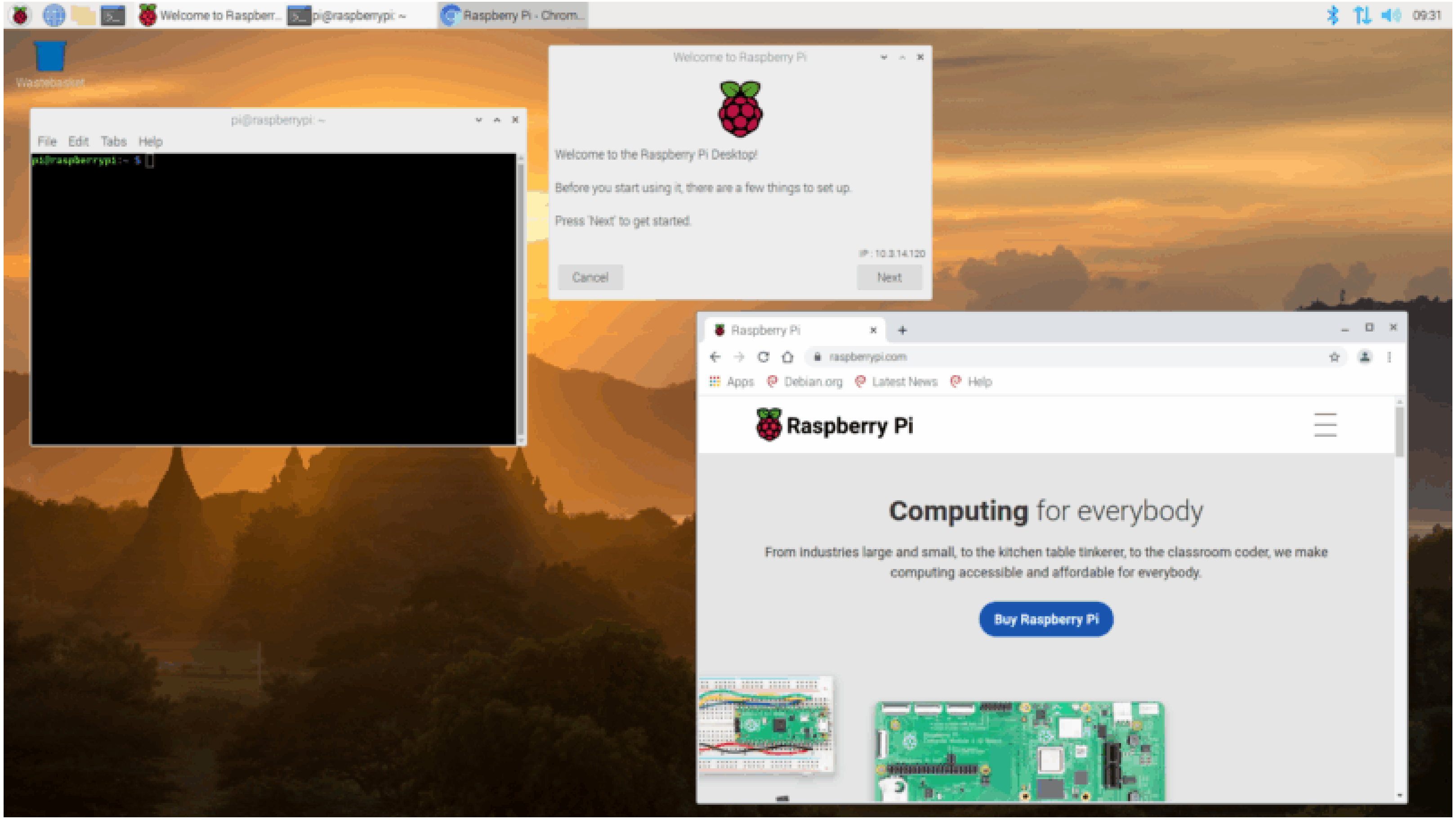


Raspberry Pi 4 – Pin Details

Category	Pins	Total Pins	Brief Explanation
Power	1, 2, 4, 17	4	Provide 3.3V and 5V supply to external components like sensors, modules, and circuits.
Ground	6, 9, 14, 20, 25, 30, 34	8	Serve as common reference (0V) for completing electrical circuits.
GPIO	7, 11, 13, 15, 16, 18, 22, 29, 31, 35, 36, 37, 38, 40	14	General-purpose pins used for **digital input/output** (e.g., LEDs, switches, relays).
I2C	3, 5, 27, 28	4	Used for **two-wire** communication (**SDA**, **SCL**) with devices like sensors and LCDs.
SPI	19, 21, 23, 24, 26	5	Supports **high-speed** communication with peripherals (ADC, DAC, displays)
UART	8, 10	2	Used for **serial communication** (TX, RX) with modules like GPS, Bluetooth.
PWM	12, 32, 33	3	**Generate pulse-width modulated signals** for controlling motors, servos, and LED brightness.

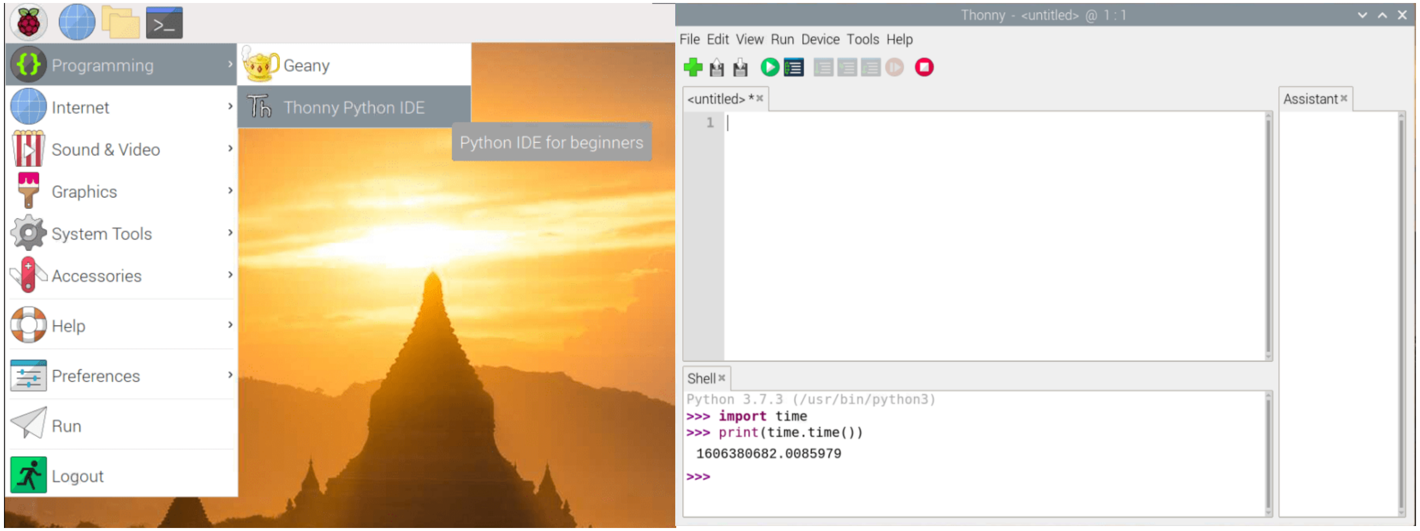
Total Pins: 40 | I2C + SPI + UART: 11 Communication Pins | GPIO + PWM: 17 Control Pins

Raspberry Pi 4 – Operating System

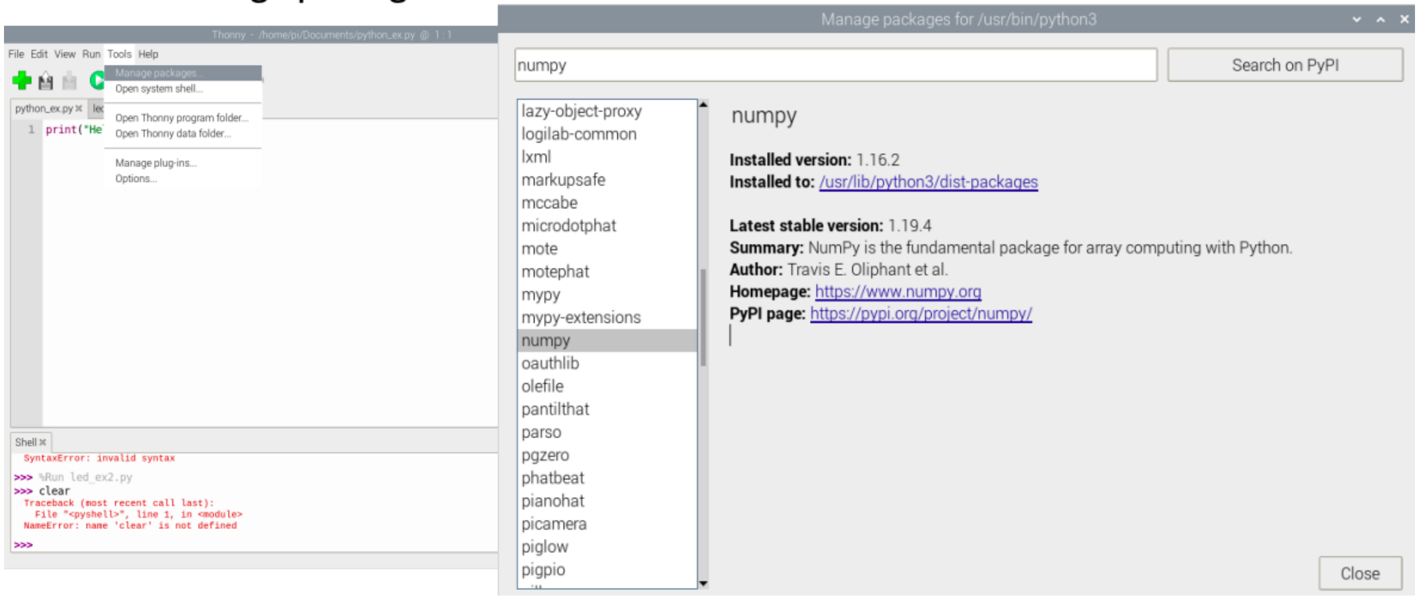


- The OS for Raspberry Pi is called Raspberry Pi OS (previously known as Raspbian)
- Raspberry Pi runs a version of an operating system called Linux (Windows and macOS are other operating systems).
- To install the necessary OS, you need a microSD card.

Raspberry Pi 4 – Operating System



Tools -> Manage packages…



- The Raspberry Pi OS comes with a basic Python Editor called Thonny
- But you can install and use other Python Editors if you prefer

https://www.raspberrypi.com/documentation/computers/os.html

Raspberry Pi 4 – GPIO Zero Useful Links

- The GPIO Zero Python Library can be used to communicate with GPIO Pins

- The GPIO Zero Python Library comes preinstalled with the Raspberry Pi OS (so no additional installation is necessary)

Resources:

- https://www.raspberrypi.org/documentation/usage/gpio/python/
- https://pypi.org/project/gpiozero/
- https://gpiozero.readthedocs.io/en/stable/
- https://gpiozero.readthedocs.io/en/stable/recipes.html

Programming Material:

https://www.halvorsen.blog/documents/technology/iot/raspberry_pi.php